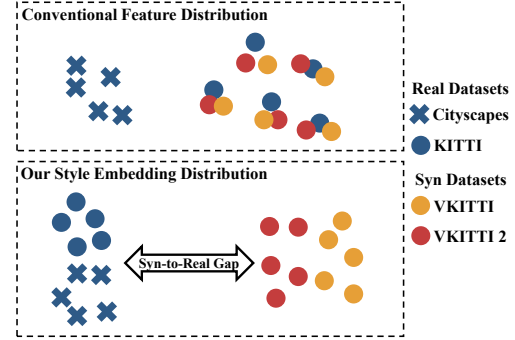


# A Style-Based Profiling Framework for Quantifying the Synthetic-to-Real Gap in Autonomous Driving Datasets

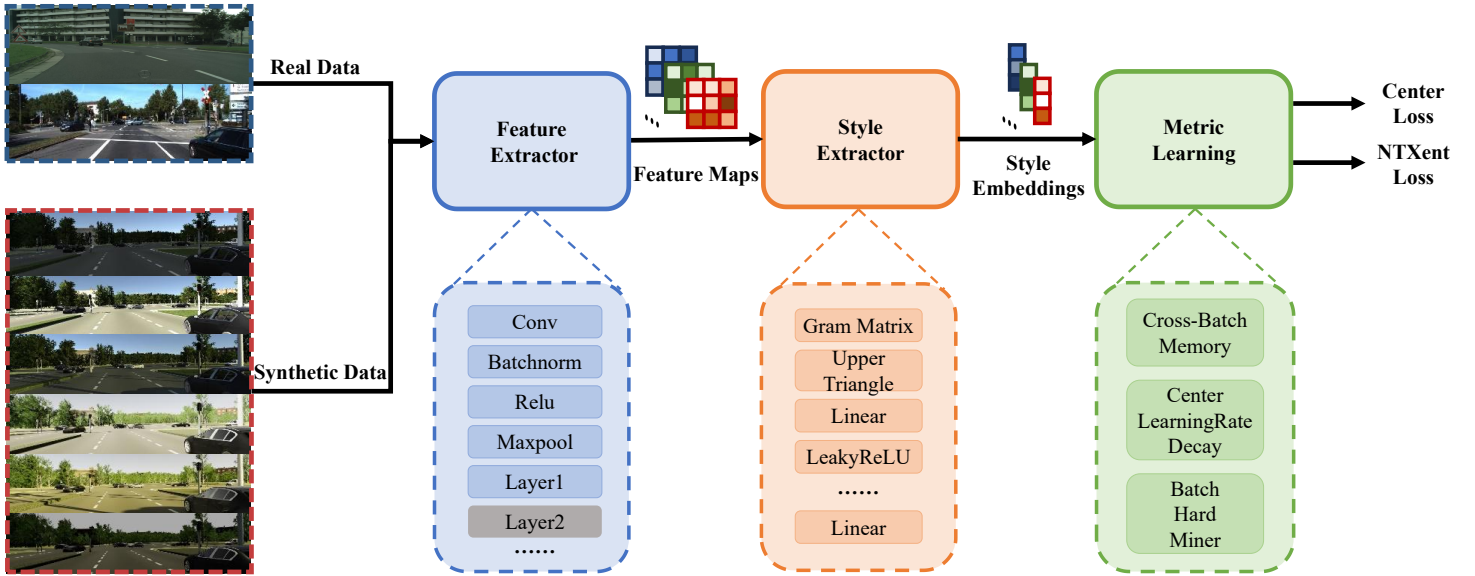
Dingyi Yao, Xinyao Han, Ruibo Ming, Zhihang Song, Lihui Peng, Jianming Hu, Danya Yao, Yi Zhang  
Department of Automation, Tsinghua University

## INTRODUCTION

- Problem:** Real-world testing for autonomous driving perception is costly, time-consuming, unsafe.
- Alternative:** Synthetic datasets offer cost-effectiveness, bias-free labeling, full scenario controllability.
- Bottleneck:** The "Synthetic-to-Real Gap" heavily limits model generalization in real-world scenarios.
- Current Metric Failures:**
  - Reference-based (PSNR/SSIM/FID):** Require paired real images (often unavailable).
  - NR-IQA (BRISQUE/NIMA):** visual quality ("how good") rather than true domain fidelity ("how realistic").
  - Prior Gap Quantification Methods for Autonomous Driving:**
    - Fail to disentangle semantic content from stylistic variations.
    - Highly sensitive to environmental factors, restricting their scope to clear weather scenarios.
- Our Contributions:**
  - A novel data-centric profiling framework to discover and characterize style profiles in datasets.
  - Introducing **Style Embedding Distribution Discrepancy (SEDD)** to quantify the Synthetic-to-Real gap.
  - Guides pre-rendering parameter screening and optimal domain adaptation model selection.



## METHODOLOGY



- Feature Extraction:** Utilizes shallow layers to capture style-related features.
- Style Extraction:** Use **Gram Matrix** to capture spatially invariant global response patterns & Reduce dimensionality.
- Metric Learning:**
  - Center Loss:** Enforces intra-class compactness.
  - NTXent Loss:** Maximizes inter-class separability with batch hard mining.
- SEDD Metrics Formulation:**
  - SEDD<sub>1</sub>:** Euclidean distance between dataset style centers.
  - SEDD<sub>2</sub>:** Maximum Mean Discrepancy (MMD) for fine-grained shifts.

$$G_{ij} = \sum_{h=1}^H \sum_{w=1}^W F_i(h, w) \cdot F_j(h, w)$$

$$\mathcal{L}_C = \frac{1}{m} \sum_{i=1}^m \|z_i - c_{y_i}\|_2 \quad l_{ij} = -\log \frac{\exp(\text{sim}(A_i, P_j)/\tau)}{\sum_{k=1}^{2m} \mathbb{I}_{[k \neq i]} \exp(\text{sim}(A_i, S_k)/\tau)} \quad \mathcal{L}_{\text{NTXent}} = \sum_{i,j} l_{ij}$$

$$\text{SEDD}_1 = \|c_{\text{new}} - c_{\text{real}}\|_2 \quad \text{SEDD}_2 = \frac{1}{m(m-1)} \sum_{i \neq j}^m k(x_i, x_j) + \frac{1}{n(n-1)} \sum_{i \neq j}^n k(y_i, y_j) - \frac{2}{mn} \sum_{i=1}^m \sum_{j=1}^n k(x_i, y_j)$$

## KEY RESULTS

- Accurate Realism Ranking:**
  - Successfully ranks dataset fidelity: KITTI > Virtual KITTI 2 > Virtual KITTI.
  - Overcomes NR-IQA biases by prioritizing objective domain realism over mere visual clarity.
- Robust Gap Quantification:**
  - Real-world datasets consistently show lower SEDD scores than synthetic.
  - Exhibits strong generalization on unseen datasets (nuScenes, SHIFT).

- Validating Sim-to-Real Methods:**
  - Quantitatively validates gap reduction for photorealism enhancements (e.g., CUT, EPE) and CARLA2Real algorithms.

***This work transforms the synthetic-to-real gap into quantifiable objective indicators, eliminating reliance on subjective experience. It provides a standardized proactive tool for the quality control of synthetic datasets in autonomous driving, facilitating more targeted improvements.***

| Datasets | Brisque<br>↓ [19] | NIMA<br>↑ [20] | Unique<br>↑ [21] | SEDD <sub>1</sub> ↓ | SEDD <sub>2</sub> ↓ |
|----------|-------------------|----------------|------------------|---------------------|---------------------|
| KITTI    | 21.6              | 3.73           | 0.11             | <b>0.102</b>        | <b>0.081</b>        |
| VKITTI   | 42.7              | <b>4.56</b>    | <b>0.13</b>      | 0.305               | 0.480               |
| VKITTI2  | <b>19.1</b>       | <b>4.56</b>    | <b>1.40</b>      | 0.274               | 0.402               |

| Datasets            | FE <sub>↓</sub> [28] | Li et al. <sup>†5</sup> [29] | Duminil et al. <sup>†100</sup> [30], [34] | Ours <sub>↓</sub>                   |
|---------------------|----------------------|------------------------------|-------------------------------------------|-------------------------------------|
|                     |                      |                              | GLCM Wavelets LBP sH                      | SEDD <sub>1</sub> SEDD <sub>2</sub> |
| KITTI [22]          | -                    | 4.845                        | 99.81 96.85 94.87 62.42                   | <b>0.103</b> <b>0.085</b>           |
| Cityscapes [27]     | 4.47                 | -                            | 96.67 98.05 87.46 72.76                   | <b>0.122</b> <b>0.111</b>           |
| nuScenes [45]       | -                    | -                            | 90.94 98.54 60.46 73.60                   | <b>0.143</b> <b>0.197</b>           |
| Virtual KITTI [7]   | 22.31                | -                            | -                                         | <b>0.316</b> <b>0.494</b>           |
| Virtual KITTI 2 [8] | -                    | -                            | 0.05 51.21 02.24 34.10                    | <b>0.275</b> <b>0.406</b>           |
| SHIFT [9]           | -                    | 4.792                        | -                                         | <b>0.210</b> <b>0.275</b>           |

| Sim-to-real Methods   | SEDD <sub>1</sub> | SEDD <sub>2</sub> |
|-----------------------|-------------------|-------------------|
| ColorTransfer [23]    | 0.242             | 0.384             |
| CUT [24]              | 0.192             | 0.278             |
| EPE2Vistas [25], [46] | 0.191             | 0.249             |
| EPE [25]              | <b>0.175</b>      | <b>0.245</b>      |
| Raw(CARLA) [47]       | 0.193             | 0.307             |
| CARLA2KITTI [26]      | <b>0.172</b>      | <b>0.158</b>      |
| CARLA2CITY [26]       | <b>0.135</b>      | <b>0.170</b>      |

